

# ARIV™ — AI Runtime Integrity Vault Module

## OOF™ Origin Open Foundation™

Independent Methodological Authority

**OriginID:** OOF-OID-AI-ARIV-2026-05-04-0001

**Category:** AI and Interpretation

**Subcategory:** Runtime Integrity & Controlled Execution

**Type:** Integrity and Validation Module

**Parent Standard:** Runtime Integrity Standard (RIS)

**Derived From:** INTEGROS® — Integrity Standard

**Version:** 1.0

**Status:** Canonical · Open Module

**Effective Date:** 4 May 2026

**Compatibility:** OOF® Methodology OS™ · Runtime Integrity Standard · INTEGROS® · TVL® · OBIDENITY™ · UCL™ · EDECI™

**Authority:** OOF®

**Protection:** MIP® — Methodological Intellectual Property

**Canonical Language:** English (UCL™)

## Canonical Definition System

**AI Runtime Integrity Vault Module (ARIV™)** defines a controlled runtime execution environment for AI systems operating within organizational or system boundaries.

ARIV™ ensures that all runtime operations involving data access, processing, output generation, and external interaction

remain:

- identity-bound
- validation-controlled
- integrity-enforced
- non-bypassable

A system is valid only when AI execution occurs within a controlled runtime environment where uncontrolled data exposure, unverified output, or unauthorized external interaction is not possible.

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## A. Module Abstract

ARIV™ establishes the runtime execution boundary for AI systems.

It prevents uncontrolled interaction with:

- internal data
- external channels
- system outputs
- execution pathways

ARIV™ transforms AI operation from an open process into a structured, auditable, and integrity-enforced runtime layer.

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## B. Core Function

ARIV™ provides:

- controlled access to system data
- validated output release
- runtime integrity enforcement
- traceability of execution behavior
- protection against uncontrolled externalization

It functions as a vault layer, ensuring that AI cannot operate outside defined execution boundaries.

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## C. System Conditions

A system implementing ARIV™ must ensure:

### 1. Identity Binding

All runtime actions are linked to a verified origin and accountable system actor under OBIDENSITY™ conditions.

### 2. Validation Enforcement

All outputs must pass validation conditions before exposure, release, execution, or downstream system use under TVL® logic.

### 3. Integrity Enforcement

Runtime conditions must remain non-bypassable and governed by INTEGROS® enforcement logic.

### 4. Controlled Execution Environment

AI may operate only within defined, monitored, and structurally bounded execution conditions.

### 5. External Communication Control

Any data, output, signal, or action leaving the runtime boundary must pass identity, validation, and integrity conditions before release.

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## D. Scope

ARIV™ applies to:

- AI-assisted enterprise systems
  - internal AI tools and workflows
  - robotics interacting with humans
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autonomous decision-support or decision-execution systems systems  
operating with sensitive, regulated, or high-risk data controlled  
enterprise or institutional AI environments

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## E. Methodology

Implementation of ARIV™ requires:

- definition of runtime execution boundaries
- classification of accessible data and system layers
- restriction of uncontrolled data exposure

mapping of all input, output, and external communication channels  
validation enforcement before any external interaction integrity  
control over runtime behavior and boundary conditions

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## F. Valid State

A system is ARIV™ valid only if:

- AI execution remains inside defined runtime boundaries
  - data access is controlled and traceable
  - outputs are validated before release
  - external interaction cannot bypass runtime controls
  - all actions remain identity-bound and auditable
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## G. Invalid Conditions

A system is considered ARIV™ invalid if:

- AI accesses data outside controlled boundaries
- outputs are generated or released without validation
- runtime actions are not traceable

external communication bypasses validation or integrity conditions  
identity binding is absent execution occurs outside monitored  
system scope

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## H. System Impact

ARIV™ transforms AI systems from:

uncontrolled tools → into controlled execution environments  
non-traceable processes → into auditable runtime systems data  
exposure risks → into integrity-governed execution workflows It  
establishes a runtime condition in which AI behavior becomes  
structurally bounded before trust is assumed.

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## I. Cross-Module Dependency

ARIV™ depends on:

- Runtime Integrity Standard (RIS) — parent runtime framework
- INTEGROS® — enforcement layer
- TVL® — validation layer
- OBIDENITY™ — identity binding layer
- EDECI™ — external communication integrity layer

ARIV™ does not replace these layers.

It enforces them at runtime.

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## J. Validation Layer Linkage

ARIV™ operates as a runtime enforcement module within the validation architecture.

Its role is to ensure that validation remains structurally enforced during execution and cannot be treated as optional postprocessing.

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## K. Minimum Implementation Framework (MIF)

### Step 1 — Define Runtime Boundary

The organization must define where AI execution begins and ends.

Minimum requirement:

- defined execution scope
- defined internal boundary
- defined external release boundary

### Step 2 — Bind Identity

All runtime actions must be attributable to a verified origin and accountable system actor.

Minimum requirement:

- runtime identity binding exists
- origin of execution is traceable
- accountable control layer is defined

### Step 3 — Control Data Access

AI must not access unrestricted data.

Minimum requirement:

- data classes are defined
  - access permissions are bounded
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- uncontrolled exposure is prevented

## Step 4 — Enforce Validation Before Release

No output may leave the runtime environment without structural validation.

Minimum requirement:

- output validation path exists
- release conditions are defined
- bypass is prevented

## Step 5 — Secure External Interaction

All outbound communication and execution pathways must remain controlled.

Minimum requirement:

- external channels are mapped
- release logic is monitored
- uncontrolled external communication is blocked

## Step 6 — Preserve Traceability

Runtime execution must remain auditable.

Minimum requirement:

- runtime actions are traceable
- system behavior is observable
- evidence can be retained and reviewed

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# L. Use Case 1 — AI Adoption Barrier in Enterprise

## Scenario

A company delays AI deployment because it does not trust uncontrolled model behavior around internal data and outputs.

## Application

ARIV™ establishes a controlled runtime vault where:

- data access is bounded
- outputs are validated
- execution remains traceable
- external release cannot bypass control

## Result

The company gains a controlled path to AI adoption instead of rejecting AI entirely.

Without such a module, organizations may avoid AI and lose significant operational value every day due to preventable trust failure.

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## M. Use Case 2 — Sensitive Internal Knowledge Environment

### Scenario

An organization wants to use AI with internal documents, workflows, and decision support, but cannot allow uncontrolled leakage or unverified output.

### Application

ARIV™ enforces:

- restricted data access
- monitored runtime conditions
- validation before output release
- identity-bound execution

## Result

AI becomes usable inside sensitive environments without turning the system into an uncontrolled exposure risk.

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## N. Use Case 3 — Robotics in Human Environments

### Scenario

A robotics system interacts with humans in real-world settings where ambiguous interpretation or uncontrolled execution creates safety and liability risk.

### Application

ARIV™ provides:

- bounded runtime behavior
- validation-controlled output and action pathways
- traceable execution conditions
- non-bypassable runtime controls

## Result

The system becomes more auditable, governable, and defensible in safety-critical environments.

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## O. Use Case 4 — Regulated AI Deployment

## Scenario

An enterprise deploys AI into a regulated environment and must show that outputs, runtime behavior, and external interaction remain controlled.

## Application

ARIV™ creates:

- controlled execution boundaries
- auditable runtime evidence
- validation-linked release logic
- integrity-enforced system behavior

## Result

The organization improves its position for audit, compliance review, client trust, and regulatory scrutiny before disputes or failures arise.

## Canonical Closing Statement

**If AI execution is not controlled, it cannot be trusted.**

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